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# SOBEL EDGE DETECTION - X AXIS

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload Image
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read Image
img = cv2.imread(filename)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Sobel X
sobel_x = cv2.Sobel(gray,
                    cv2.CV_64F,
                    1, # dx
                    0, # dy
                    ksize=3)

sobel_x = cv2.convertScaleAbs(sobel_x)

# Display
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(sobel_x, cmap='gray')
plt.title("Sobel X")
plt.axis("off")

plt.show()
```

[Choose Files](#) building image.jpg
building image.jpg(image/jpeg) - 40981 bytes, last modified: 10/8/2026 - 100% done
Saving building image.jpg to building image.jpg

Original Image



Sobel X

